

X1
DigitalCarrierJack.SchDoc

Encoder1
Encoder1.SchDoc

Encoder2
Encoder2.SchDoc

Encoder3
Encoder3.SchDoc

Connector
Connectors.SchDoc

DIG_IO
PowerSupply
DIG_Res
Status
I2C
CPLD_IO

DIG_IO
PowerSupply
DIG_Res
Status
I2C
CPLD_IO

Encoder_Signals_1
Encoder_Signals_2
Encoder_Signals_3

Encoder_Signals

Encoder_Signals

Encoder_Signals

GND
GND



Revision

Serial
Serial

UltraZohm_DigitalVoltage
Project

Designer

Date

Title		TopSheet.SchDoc		Technische Hochschule Nürnberg Georg Simon Ohm Fürther Straße 250 90429 Nürnberg GERMANY			
Size:	A4	Revision:	1v00				
Date:	07.02.2022	Time:	15:33:58	Sheet 1	of 6		TECHNISCHE HOCHSCHULE NÜRNBERG GEORG SIMON OHM
Project:	UltraZohm_Digital_Incremental_Encoder_v2.PrjPCB						
					Author:	Michaela Hlatky	

A

A

B

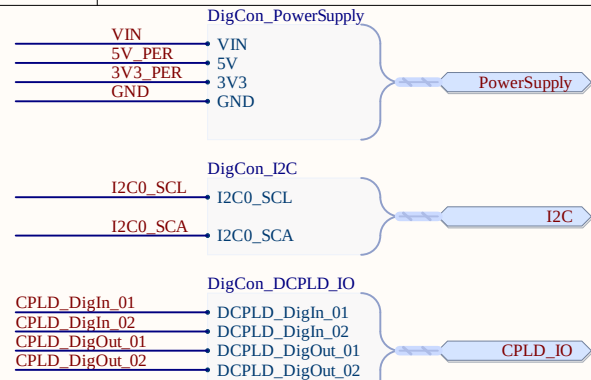
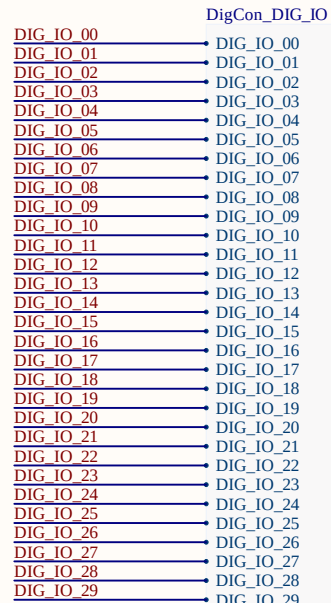
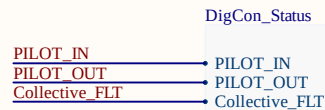
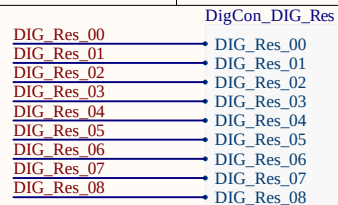
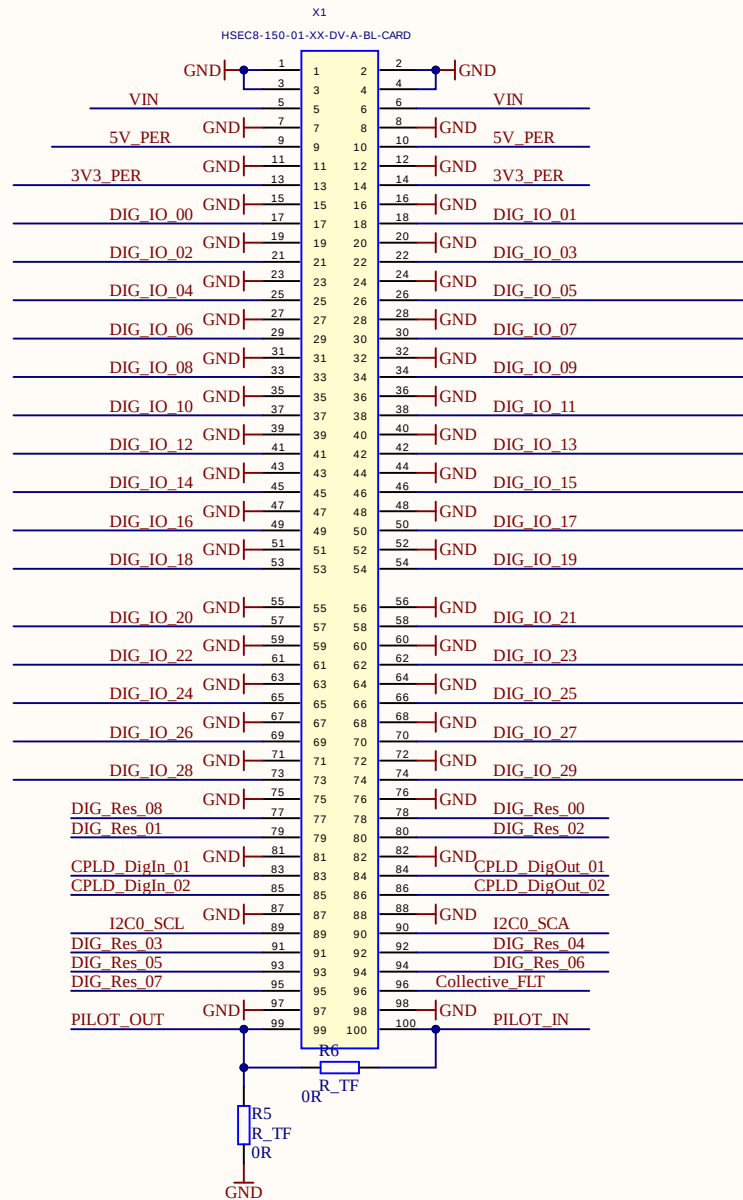
B

C

C

D

D



A

A

B

B

C

C

D

D

DigCon_DIG_IO

DIG_IO_00 → DIG_IO_00
DIG_IO_01 → DIG_IO_01
DIG_IO_02 → DIG_IO_02
DIG_IO_03 → DIG_IO_03
DIG_IO_04 → DIG_IO_04
DIG_IO_05 → DIG_IO_05
DIG_IO_06 → DIG_IO_06
DIG_IO_07 → DIG_IO_07
DIG_IO_08 → DIG_IO_08
DIG_IO_09 → DIG_IO_09
DIG_IO_10 → DIG_IO_10
DIG_IO_11 → DIG_IO_11
DIG_IO_12 → DIG_IO_12
DIG_IO_13 → DIG_IO_13
DIG_IO_14 → DIG_IO_14
DIG_IO_15 → DIG_IO_15
DIG_IO_16 → DIG_IO_16
DIG_IO_17 → DIG_IO_17
DIG_IO_18 → DIG_IO_18
DIG_IO_19 → DIG_IO_19
DIG_IO_20 → DIG_IO_20
DIG_IO_21 → DIG_IO_21
DIG_IO_22 → DIG_IO_22
DIG_IO_23 → DIG_IO_23
DIG_IO_24 → DIG_IO_24
DIG_IO_25 → DIG_IO_25
DIG_IO_26 → DIG_IO_26
DIG_IO_27 → DIG_IO_27
DIG_IO_28 → DIG_IO_28
DIG_IO_29 → DIG_IO_29

DIG_IO

Encoder_Signals_CH3

A_encoder → A_encoder3
B_encoder → B_encoder3
Z_encoder → Z_encoder3

Encoder_Signals_3

Encoder_Signals_CH2

A_encoder → A_encoder2
B_encoder → B_encoder2
Z_encoder → Z_encoder2

Encoder_Signals_2

Encoder_Signals_CH1

A_encoder → A_encoder1
B_encoder → B_encoder1
Z_encoder → Z_encoder1

Encoder_Signals_1

DIG_IO_00 01
DIG_IO_01 04
DIG_IO_03 07
DIG_IO_05 10
DIG_IO_07 13
DIG_IO_09 16
DIG_IO_11 19
DIG_IO_13 22
DIG_IO_15 25
DIG_IO_17 28
01 02
04 05
07 08
10 11
13 14
16 17
19 20
22 23
25 26
28 29
02 DIG_IO_19
05 DIG_IO_02
08 DIG_IO_04
11 DIG_IO_06
14 DIG_IO_08
17 DIG_IO_10
20 DIG_IO_12
23 DIG_IO_14
26 DIG_IO_16
29 DIG_IO_18

GND

X2

03 03
06 06
09 09
12 12
15 15
18 18
21 21
24 24
27 27
30 30

Pin Header

VIN_fused 01
GND 03
5V_PER_fused 05
GND 07
3V3_PER_fused 09
GND 11
01 02
03 04
05 06
07 08
09 10
11 12
02 VIN_fused
04 GND
06 5V_PER_fused
08 GND
10 3V3_PER_fused
12 GND

Pin Header

DigCon_DIG_Res

DIG_Res_00 → DIG_Res_00
DIG_Res_01 → DIG_Res_01
DIG_Res_02 → DIG_Res_02
DIG_Res_03 → DIG_Res_03
DIG_Res_04 → DIG_Res_04
DIG_Res_05 → DIG_Res_05
DIG_Res_06 → DIG_Res_06
DIG_Res_07 → DIG_Res_07
DIG_Res_08 → DIG_Res_08

DIG_Res

DigCon_DCPLD_IO

DCPLD_DigIn_01 → CPLD_DigIn_01
DCPLD_DigIn_02 → CPLD_DigIn_02
DCPLD_DigOut_01 → CPLD_DigOut_01
DCPLD_DigOut_02 → CPLD_DigOut_02

CPLD_IO

DigCon_Status

PILOT_IN → PILOT_IN
PILOT_OUT → PILOT_OUT
Collective_FLT → Collective_FLT

Status

DIG_IO_21 01
DIG_IO_23 04
DIG_IO_25 07
DIG_IO_27 10
DIG_IO_29 13
DIG_Res_07 16
CPLD_DigIn_01 19
CPLD_DigIn_02 22
I2C0_SCL 25
PILOT_OUT 28
01 02
04 05
07 08
10 11
13 14
16 17
19 20
22 23
25 26
28 29
02 DIG_IO_20
05 DIG_IO_22
08 DIG_IO_24
11 DIG_IO_26
14 DIG_IO_28
17 DIG_Res_08
20 CPLD_DigOut_01
23 CPLD_DigOut_02
26 I2C0_SCA
29 PILOT_IN

GND

X3

03 03
06 06
09 09
12 12
15 15
18 18
21 21
24 24
27 27
30 30

Pin Header

VIN → VIN_fused
Littelfuse_PTC_1812
5V_PER → 5V_PER_fused
Littelfuse_PTC_1812
3V3_PER → 3V3_PER_fused
Littelfuse_PTC_1812

DigCon_I2C

I2C0_SCL → I2C0_SCL
I2C0_SCA → I2C0_SCA

I2C

DigCon_PowerSupply

VIN → VIN
5V → 5V_PER
3V3 → 3V3_PER
GND → GND

PowerSupply

Title Connectors.SchDoc

Size: A4

Revision: 1v00

Date: 07.02.2022 Time: 15:33:58

Sheet 3 of 6

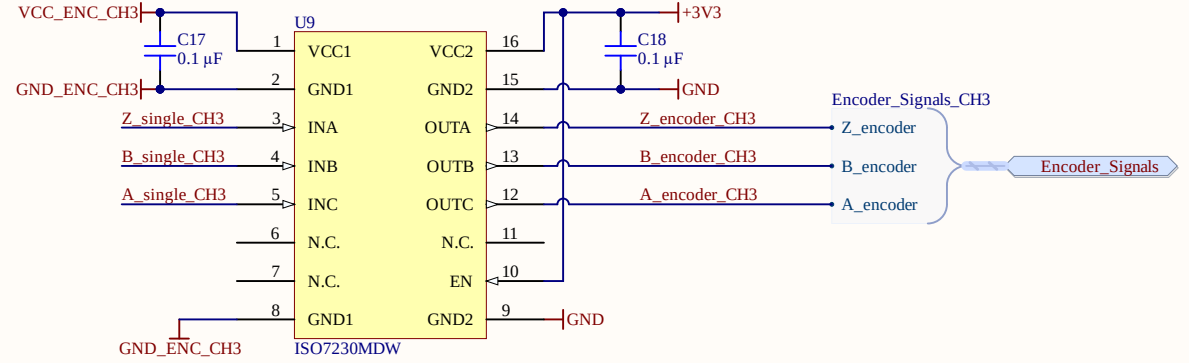
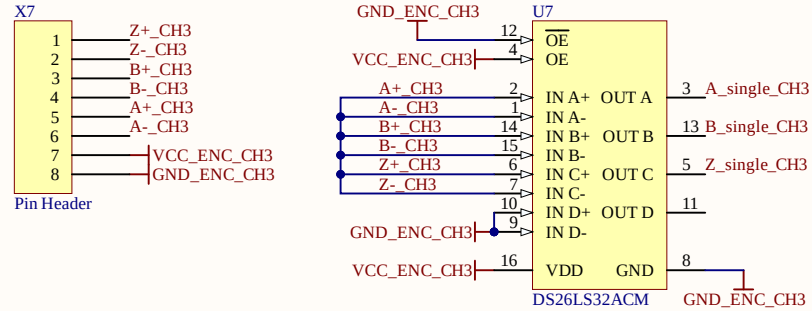
Author: Michaela Hlatky

Project: UltraZohm_Digital_Incremental_Encoder_v2.PrjPCB

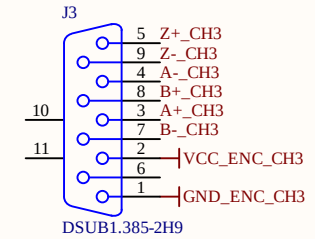
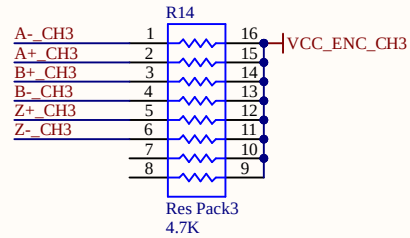
Technische Hochschule Nürnberg Georg Simon Ohm
Fürther Straße 250
90429 Nürnberg
GERMANY



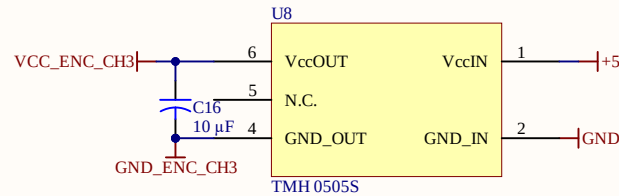
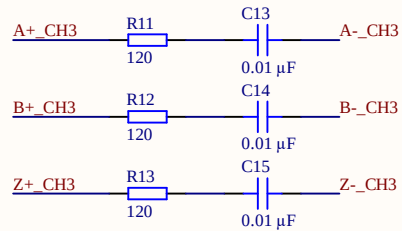
A



B

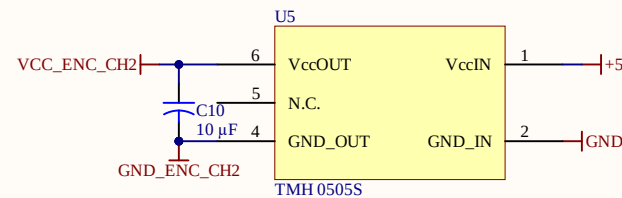
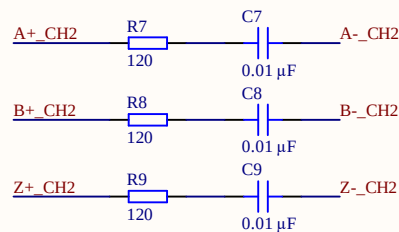
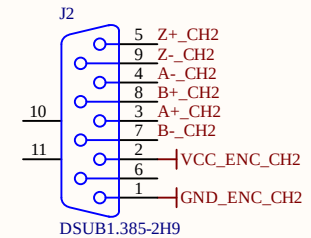
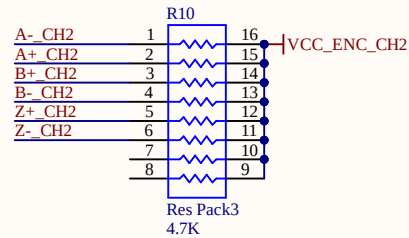
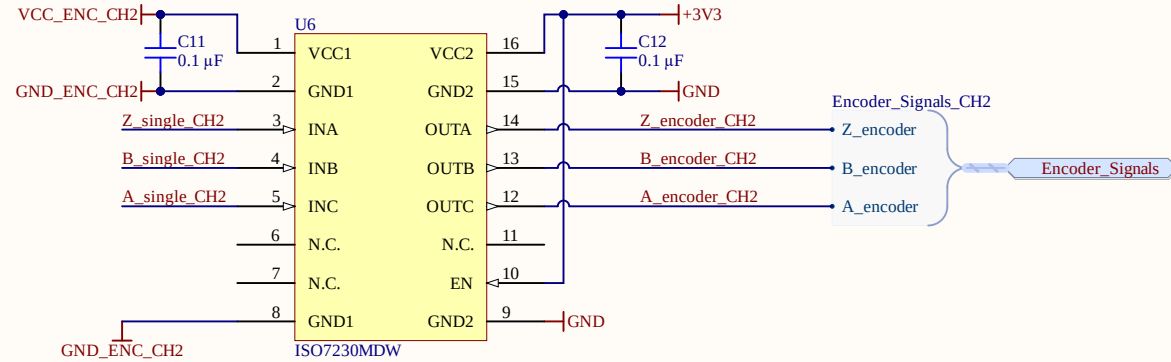
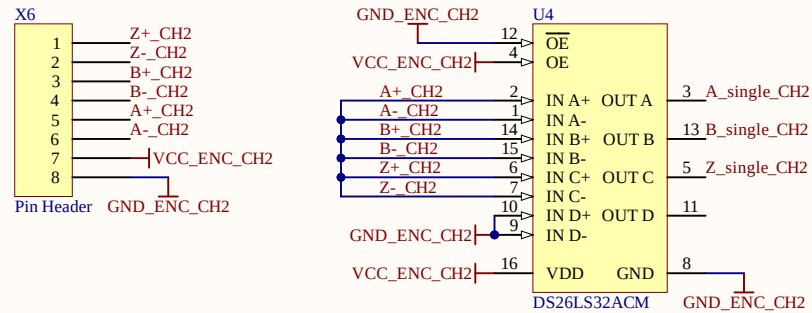


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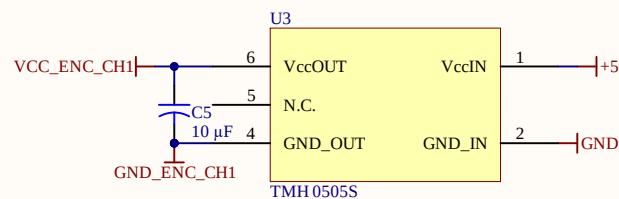
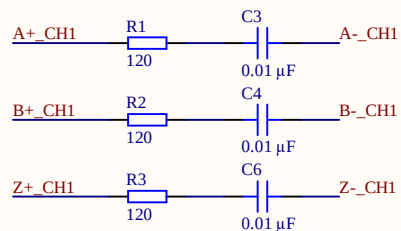
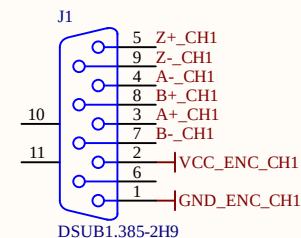
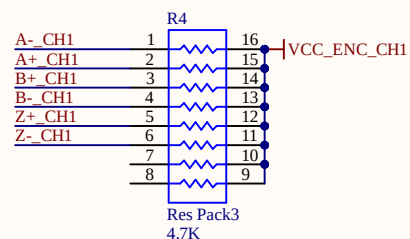
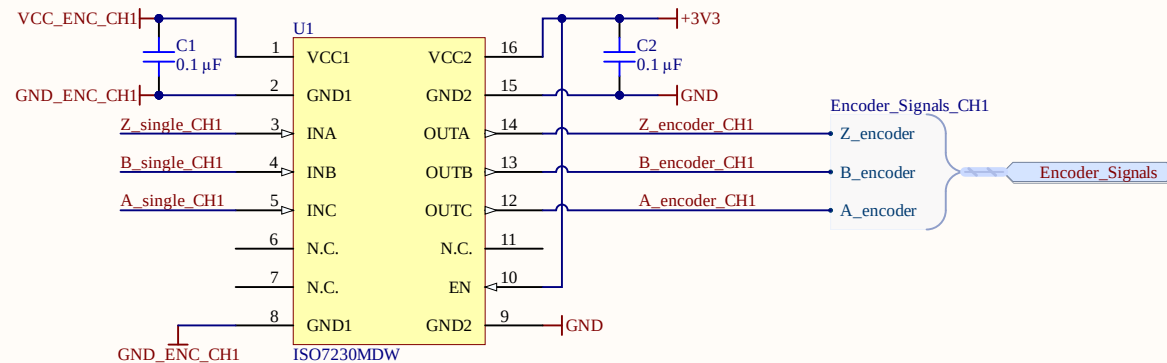
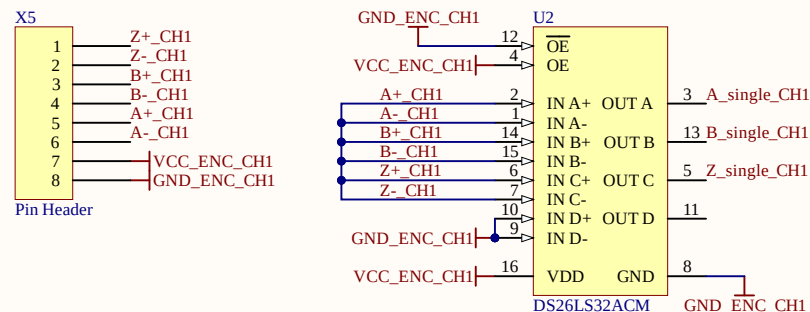


Label	Pin	Encoder	Color(Kübler)
U0+	Pin3	0+	blue
U0-	Pin4	0-	red
U1+	Pin8	A+	green
U1-	Pin7	A-	yellow
U2+	Pin5	B+	grey
U2-	Pin9	B-	pink
VCC_ENC	Pin2	UB	brown
GND_ENC	Pin1	0V	white
NC	Pin6	--	

D



Label	Pin	Encoder	Color(Kübler)
U0+	Pin3	0+	blue
U0-	Pin4	0-	red
U1+	Pin8	A+	green
U1-	Pin7	A-	yellow
U2+	Pin5	B+	grey
U2-	Pin9	B-	pink
VCC_ENC	Pin2	UB	brown
GND_ENC	Pin1	0V	white
NC	Pin6	--	



Label	Pin	Encoder	Color(Kübler)
U0+	Pin3	0+	blue
U0-	Pin4	0-	red
U1+	Pin8	A+	green
U1-	Pin7	A-	yellow
U2+	Pin5	B+	grey
U2-	Pin9	B-	pink
VCC_ENC	Pin2	UB	brown
GND_ENC	Pin1	0V	white
NC	Pin6	--	